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The A1C Test & Race/Ethnicity

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What is the A1C test?

The

A1C test

is a blood test that provides information about your average levels of blood glucose

, also called blood sugar, over the past 3 months. The test is also called hemoglobin A1C, HbA1C, and glycated hemoglobin, and it is the blood glucose test that doctors most often use for managing diabetes

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How the test works

The A1C test measures the amount of glucose in your blood that has become attached to hemoglobin

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protein

that allows red blood cells to carry oxygen from your lungs to the rest of your body. When glucose enters your blood, it binds with the hemoglobin in your red blood cells. The more glucose in your blood, and the longer it stays elevated, the more it will attach to the hemoglobin.

The A1C test result is reported as a percentage. The higher the percentage, the higher your blood glucose levels. A normal A1C level is below 5.7 percent.

The A1C blood test reflects your average blood glucose levels over the past 3 months.

How can my racial/ethnic background affect the results of my A1C test?

The form of hemoglobin in your blood depends on genes you inherit from your parents. Many different forms of hemoglobin exist. The most common form is hemoglobin A, also called normal adult hemoglobin. Other, less common forms of hemoglobin are called hemoglobin variants.

Hemoglobin variants can affect the results of some A1C tests. These variants are more common in some parts of the world, such as Africa, South and Southeast Asia, and the Mediterranean. If your ancestors came from these regions, you may have a form of hemoglobin in your blood that could affect your A1C test results.

Depending on your descent, you could have a hemoglobin variant that can affect A1C test results.

What are some common hemoglobin variants?

The most common hemoglobin variants that can affect A1C test results are hemoglobin S, C, D, and E.

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These variants can also cause hemoglobinopathies, a group of blood disorders and diseases that affect red blood cells.

Table 1 lists each variant, the racial and ethnic groups most likely to have the variant, and the blood disorders the variant can cause. In most cases, you will only develop these health problems if you inherit genes for hemoglobin variants from both of your parents. If you inherit only one hemoglobin variant, along with normal hemoglobin, you usually won't develop any symptoms or health problems.

Table 1. Common hemoglobin variants that can affect A1C test results

Hemoglobin Variant